

Vulnerability Assessment Report

1st January 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is very valuable because it manages large amounts of data, customer data among the different kinds of data and a critical consideration. Many remote employees access this data that it's open to the public, data that might contain PII or SPII. It's important for the business to secure the data on the server because threat sources with malicious intent could try to get a hold of this information that is used for regular marketing operations. If the server was disabled, there would be a negative impact on the organization, being a critical server accessed continuously it would most likely cause a stop on operations which would lead to a financial loss. The purpose of this vulnerability assessment is to protect the company's critical server and stored information from getting leaked, altered or destroyed.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9
Hacktivist	Conduct a Denial of Service (DoS) attack	2	3	6

<i>Customer</i>	<i>Alter/Delete critical information</i>	<i>2</i>	<i>3</i>	<i>6</i>
<i>Competitor</i>	<i>Disrupt mission-critical operations.</i>	<i>1</i>	<i>3</i>	<i>3</i>

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs. The database has been open to the public for three years, these four specific threat sources/events are examples of what might happen if security controls are not applied on the server containing sensitive information.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS 1.2 or higher, instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database. Following the principle of least privilege will deny access to any unauthorized user.